

RED TEAM ENGAGEMENT REPORT

Executive Summary:

A simulated **red team** engagement was conducted on a controlled lab environment consisting of **kali Linux (attacker), Metasploitable2/3 (victims), and Windows 10 VM**. The engagement successfully demonstrated critical vulnerabilities including **remote Code Execution, credential dumping, lateral movement, and data exfiltration via DNS tunneling**.Risk Level: **CRITICAL**

Findings:

	Vulnerability	CVSS	System	Impact
1	Samba RCE	9.8	Metasploitable2	Critical
2	SMB PSEXEC	9.0	Metasploitable3	Critical
3	NTLM Hash Dump	8.5	Metasploitable3	High
4	Phishing	7.5	Windows VM	High
5	Buffer Overflow	9.8	Vulnerable Binary	Critical
6	DNS Tunneling	7.0	Metasploitable3	High

Recommendations:

1. Update Samba to the latest version.
2. Block SMB ports 139 and 445 through the firewall.
3. Implement a strong password policy.
4. Disable NTLM authentication.

5. Monitor DNS traffic.
6. Implement email filtering.
7. Deploy a patch management system.
8. Deploy an Endpoint Detection and Response (EDR) solution.

Attack Flowchart



